

SAMUEL SETSOFIA

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EDUCATION

Northwestern University

MSc Artificial Intelligence

Evanston, IL; Expected December 2025

- ML Systems Architecture, Cloud-Native AI Solutions, Multi-Agent Systems, Production ML Deployment, ML Ops

University of Ghana

BSc Biomedical Engineering

Accra, Ghana; July 2020

RELEVANT EXPERIENCE

CodebuddyAI, AI Solutions Architect

Chicago, IL; June 2025 - September 2025

- Built enterprise RAG infrastructure leveraging Amazon Titan Embeddings v2 (1536D), PostgreSQL pgvector, and hybrid search capabilities with real-time query generation
- Designed and deployed AWS cloud-native infrastructure (EC2, Bedrock, Aurora, Lambda) managing 100k+ daily API calls at 99.9% uptime while reducing operational costs by 50%
- Accelerated development velocity by 70% using AI-assisted tools (Cursor, Bolt, Cline, Claude), delivering production-ready MVP in 14 days

Northwestern University, Assistant Learning Engineer

Evanston, IL; September 2024 - May 2025

- Co-led university-wide AI adoption initiative impacting 5,000+ students and 100+ faculty, establishing Northwestern as leader in AI education
- Architected cloud-based training infrastructure supporting concurrent workshops for 200+ participants with real-time AI model deployment
- Designed suite of AI-powered teaching tools for improving faculty AI tools adoption by 60%

IRC WASHNote, AI Engineer & Technical Product Manager

Rotterdam, Netherlands; October 2023 - August 2024

- Engineered document processing pipeline for 1,000+ research papers achieving 93% accuracy for 500+ international development professionals
- Established hybrid retrieval-augmented generation system using vector databases (Pinecone, Chroma DB), improving information retrieval speed by 10x - 1000%
- Built Neo4j-based knowledge graph system connecting 50,000+ entities, enabling advanced reasoning and reducing research time by 67%

CopiantoAI, Chief Technology Officer

Accra, Ghana; May 2022 - September 2023

- Implemented production REST APIs with Django Rest Framework supporting high-volume transactions
- Integrated analytics and customer engagement APIs across 4 sources (Google Analytics, Hotjar, Mixpanel, Crisp)
- Constructed web scraping solutions using Scrapy for data extraction of over 500+ sources and preprocessing at scale

KEY SKILLS

ML Architecture	RAG Agentic RAG LLM Integration Ensemble Methods Agentic Frameworks ML Ops
Cloud Services	AWS (EC2, Bedrock, Aurora, Lambda, CloudFormation) GCP Azure Auto-Scaling Microservices
Data Systems	PostgreSQL pgvector Pinecone Chroma DB Neo4j MongoDB Elasticsearch Hybrid Search
Dev Stack	Python Django FastAPI LangChain LangGraph TensorFlow PyTorch Docker Kubernetes
AI Tools	Cursor Claude Code Hugging Face Llama Index Amazon Titan Embeddings AWS Bedrock